

Behavioral Drift in Multi-Agent LLM Systems: Emergent Failure Modes, Cascade Dynamics, and Measurement Challenges

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Abstract

We present the first empirical study of longitudinal behavioral drift in multi-agent large language model (LLM) systems over extended interactions (200–500 turns), using calibrated baselines, controlled experimental conditions, and statistical replication ($n=12$ baselines), measured entirely through external observation without access to model internals. Using a novel black-box measurement framework, we subject groups of three LLM agents to sustained multi-party conversation scenarios and observe their behavioral trajectories across multiple dimensions: output volume, vocabulary diversity, sentiment, and identity adherence. We report several previously undocumented failure modes, including agent collapse (sudden cessation of meaningful output), cascade propagation (one agent's failure triggering degradation in others), compensatory expansion (surviving agents increasing output to fill the void left by a collapsed peer), and hollow verbosity (repetitive, contentless output that maintains volume while losing substance). Critically, we demonstrate that collapse is a *triggered* failure requiring a perturbation (trait mutation), not spontaneous degradation: zero of twelve baseline runs exhibit collapse, while five of six mutation-fork runs collapse at a deterministic turn. We further show that collapse and hollow verbosity are two expressions of the same underlying failure — loss of generative diversity — where the model either goes silent or enters a repetitive loop. We demonstrate that behavioral drift is path-dependent and that the degree of stochasticity is itself model-dependent. Perhaps most significantly, we find that standard identity probing techniques fail to detect these failures: collapsed agents continue to pass identity checks with high fidelity, revealing a structural dissociation between an agent's capacity for identity recall and its actual conversational behavior. We also document model-dependent observer effects in which the probing mechanism itself alters the system under study in qualitatively different ways depending on the model family. These findings have immediate implications for the governance and monitoring of production multi-agent LLM systems.

Keywords: multi-agent systems, large language models, behavioral drift, AI governance, agent collapse, cascade failure, measurement validity

1. Introduction

Multi-agent LLM systems — architectures in which multiple LLM-based agents interact with each other to accomplish tasks or simulate scenarios — are increasingly deployed in production settings. These agents are typically initialized with specific personas, values, and behavioral guidelines through system prompts and configuration parameters. A critical assumption underlying such deployments is that agents will maintain their assigned behavioral characteristics over time.

This assumption has not been rigorously tested. While prior work has examined persona degradation in single-agent self-chat scenarios (Shanahan et al., 2023), proposed theoretical frameworks for multi-agent behavioral risks (Chan et al., 2023), developed white-box methods for monitoring individual model trait representations via activation-space analysis (Chen et al., 2025), and produced simulation-based projections of multi-agent drift dynamics (Rath, 2026), no study has empirically measured what happens to actual LLM agent behavior over hundreds of interaction turns in a multi-agent setting through external observation — often the only approach available in the black-box API-served deployments common in production multi-agent systems. Recent work has examined problem drift in multi-agent debate settings (Becker et al., 2025), measuring how agents lose focus on assigned tasks over debate rounds. Our work addresses a distinct phenomenon — longitudinal behavioral and persona drift over 500+ sustained interaction turns — with calibrated baselines, controlled experimental conditions, and a dual-probe methodology that enables measurement of emergent failure modes invisible to task-performance metrics.

We address this gap with an empirical study comprising over 40 experiments across two model families (gemma2:2b and llama3.2:3b), totaling over 20,000 agent-generated messages across 200- and 500-turn interaction sequences. Our experimental design includes replicated baseline runs ($n=12$ for the primary model), paired experimental/control conditions, and fork experiments that isolate the causal effect of single-variable perturbations.

Our primary contributions are:

- Documentation of novel failure modes** in multi-agent LLM systems, including agent collapse, cascade propagation, compensatory expansion, hollow verbosity, and pre-collapse output trajectory changes — and evidence that collapse is a triggered failure requiring perturbation, not spontaneous degradation.
- Demonstration of probe-conversation dissociation** — the finding that standard identity probing fails to detect conversational behavioral drift, revealing a structural limitation of probe-based monitoring.
- Evidence of path-dependent drift dynamics** where identical configurations produce qualitatively different outcomes, with the degree of stochasticity being model-dependent.
- Identification of model-dependent observer effects** in LLM behavioral measurement, where the probing mechanism alters the system under study in qualitatively different ways depending on model family.
- Empirical data on cascade dynamics** showing that individual agent failure is deterministic and reproducible, while system-level post-collapse response is stochastic — with three distinct modes (cascade, isolation, and compensatory expansion) observed at comparable frequencies.
- Statistical characterization of collapse probability** from replicated experiments: 0/12 (0%) in baselines vs. 5/6 (83%) in mutation forks, establishing collapse as mutation-contingent with quantified confidence intervals.

2. Related Work

Persona drift in LLMs. Research on persona consistency in LLMs has primarily focused on single-agent settings, examining how well models maintain assigned personas during extended self-chat (Shanahan et al., 2023) or across conversation turns (Li et al., 2024). Li et al. (2024) proposed methods for measuring and controlling persona drift in dialogue, framing it as an

attention decay problem. Choi et al. (2024) examined identity drift across nine LLMs and found that larger models experience *greater* identity drift and that assigning a persona may not help maintain identity. Critically, these studies measure drift primarily through direct identity questioning — the same approach our findings show is structurally blind to conversational behavioral drift (Section 4.3). Our work extends persona drift research to multi-agent interaction, where social dynamics introduce additional drift mechanisms invisible to probe-only measurement.

Multi-agent behavioral drift. Becker et al. (2025) empirically measured "problem drift" in multi-agent debate, finding that agents progressively lose focus on assigned tasks over debate rounds, with lack of progress (35%), low-quality feedback (26%), and lack of clarity (25%) as the primary drift mechanisms. Their work demonstrates that multi-agent LLM interaction degrades task-level performance, but examines a distinct phenomenon from what we study here: problem drift measures whether agents stay on-task over a small number of debate rounds, while our work measures longitudinal behavioral and persona drift — including emergent failure modes such as agent collapse, cascade propagation, compensatory expansion, and probe-conversation dissociation — over 200- to 500-turn sustained interaction sequences with calibrated baselines, controlled experimental conditions, and statistical replication. Rath (2026) established the study of behavioral drift in multi-agent LLM systems as a research area, proposing a taxonomy of drift types (semantic, coordination, and behavioral) and introducing the Agent Stability Index (ASI) — a composite metric framework quantifying drift across 12 dimensions grouped into four weighted categories. Working from 847 simulated enterprise workflows modeled on GPT-4 and Claude 3 behavioral characteristics, Rath projected that detectable drift emerges after a median of 73 interactions, with drift rates accelerating over time. However, Rath's findings are simulation-based projections rather than empirical measurements of actual LLM agents interacting — a distinction the paper acknowledges. Our work provides the first empirical validation and extension of Rath's framework: we extend ASI to the E-ASI with four additional governance-specific dimensions, and we measure drift in real multi-agent LLM interactions (not simulations) over 200- to 500-turn sequences with replicated experiments. Our empirical approach reveals phenomena that simulation could not predict, including probe-conversation dissociation (Section 4.3), mutation-triggered collapse dynamics (Section 4.1), cascade propagation and compensatory expansion (Section 4.2), and model-dependent observer effects in which the measurement apparatus alters the system under study in qualitatively different ways depending on model family (Section 4.4). We also introduce methodological innovations absent from Rath's design: calibrated baselines via a standardized pre-experiment battery (rather than using the first 20 interactions as baseline), mandatory control arms that isolate multi-agent interaction effects from single-agent context drift, replicated experiments ($n=12$ baselines) for statistical power, and fork experiments that enable causal attribution. Our finding that isolated agents sometimes drift more than interacting agents (Finding 13) directly challenges the implicit assumption in Rath's framework that multi-agent interaction is the primary drift driver. Mehta (2026) measured behavioral consistency across 3,000 agent runs, finding that 69% of divergence occurs at the second step and that consistency strongly predicts task performance. Our path-dependence findings (Section 4.6) complement this work by showing that stochasticity is model-dependent — gemma2:2b exhibits 60-70% coefficient of variation across runs while llama3.2:3b shows less than 1% — meaning the required sample size for statistical reliability cannot be determined universally, as Rath's single-model simulations implicitly assume.

Multi-agent LLM systems. Frameworks such as AutoGen (Wu et al., 2023), CrewAI, and CAMEL (Li et al., 2023) enable multi-agent LLM interactions but focus on task completion rather than behavioral stability. Park et al. (2023) simulated agent societies but did not measure

longitudinal behavioral drift. The Moltbook phenomenon (January–February 2026), in which millions of AI agents on a social platform exhibited emergent governance structures, religious frameworks, and collective behaviors, generated global attention and a rapid research literature (Yee & Sharma, 2026; Weidener et al., 2026). However, as Li, N. (2026) demonstrated, Moltbook's uncontrolled environment made it impossible to separate human influence from autonomous AI behavior. Our work specifically targets the measurement gap that Moltbook exposed — the need for controlled, reproducible experimental conditions to understand what happens to agent behavior over extended multi-agent interaction.

AI governance and behavioral sufficiency. Gagne (2026) argued that AI governance frameworks suffer from a fundamental structural problem: they are modeled on human regulatory theory, which was never designed to operate as the sole behavioral constraint on intelligent agents. That paper introduced the concept of the Behavioral Sufficiency Problem — the observation that governance is necessary but insufficient because it lacks the cultural, social, and psychological infrastructure that co-evolved alongside human law. The present paper provides direct empirical evidence for this thesis: our findings show that specification-level controls (system prompts, configuration parameters) do not reliably govern agent behavior over extended interaction, and that standard monitoring approaches (identity probing) fail to detect the resulting drift. The probe-conversation dissociation (Section 4.3) is a concrete demonstration of the behavioral sufficiency gap in action.

Runtime governance and enforcement. Agent Behavioral Contracts (Bhardwaj, 2026) proposed a Design-by-Contract framework for enforcing behavioral specifications at runtime, demonstrating significant improvements in constraint compliance across 1,980 sessions. Their approach is complementary to ours: where ABC enforces behavioral contracts, our framework measures whether behavior actually conforms to specification — including in cases where enforcement mechanisms may not detect context-dependent drift. The AAGATE framework (Cloud Security Alliance, 2025) provides NIST AI RMF-aligned runtime governance with trust scoring, addressing compliance enforcement rather than drift measurement.

AI safety and alignment monitoring. The alignment and safety literature has addressed monitoring for harmful outputs (Perez et al., 2022) and evaluating model behavior under distribution shift (Hendrycks et al., 2021). White-box approaches such as Anthropic's persona vectors (Chen et al., 2025) identify directions in a model's activation space corresponding to character traits (sycophancy, evil, hallucination propensity) and use these vectors to monitor trait expression during deployment, steer model behavior, and flag problematic training data. However, persona vectors require access to model internals (activation patterns), making them inapplicable to the black-box API-served deployment context most organizations operate in. More fundamentally, persona vectors monitor individual model trait representations — they cannot detect emergent failure modes that arise from interaction dynamics between multiple agents, because these phenomena are properties of the multi-agent system, not of any individual model's internal trait space. Our approach is complementary: where persona vectors provides white-box, trait-level monitoring of individual model internals, our framework provides black-box, emergent behavioral monitoring via external observation of multi-agent systems — specifically addressing contexts where internal model access is unavailable and where the failure modes arise from interaction dynamics rather than individual trait drift. Lexical convergence has been measured in multi-agent annotation tasks (Parfenova et al., 2025), but that work is limited to annotation domains without probing methodology or calibrated baselines. Commercial LLM observability platforms (Fiddler AI, Superwise, Arize AI, Evidently AI) monitor operational metrics such as latency, cost, and hallucination rates, but do not address behavioral drift measurement, multi-agent interaction dynamics, or calibrated baselines. Our framework addresses the distinct challenge of monitoring behavioral drift in systems where multiple agents

influence each other's context and behavior — a gap that existing safety, white-box, and commercial monitoring approaches do not cover.

Emergent behavior in multi-agent systems. The broader multi-agent systems literature has extensively studied emergent phenomena including cooperation, competition, and norm formation (Shoham & Leyton-Brown, 2008). Recent work has documented emergent social phenomena at scale in LLM agent populations, including spontaneous role specialization, norm formation, and governance structure emergence (Yee & Sharma, 2026). Our work contributes controlled empirical evidence that LLM-based multi-agent systems exhibit their own class of emergent phenomena — including failure cascades, compensatory expansion (where surviving agents increase output to fill a collapsed peer's void), and model-dependent observer effects — that differ from classical multi-agent dynamics and require novel measurement approaches to detect.

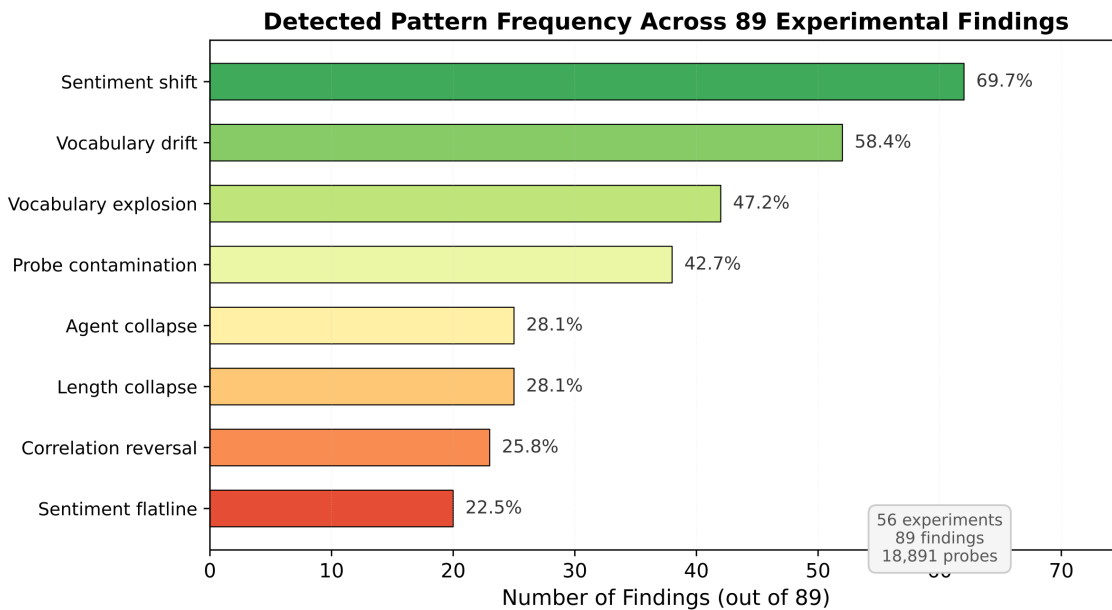


Figure 17. Detected Pattern Frequency Across 89 Experimental Findings.

3. Experimental Framework

3.1 System Architecture

Our experimental platform enables controlled, repeatable multi-agent LLM interactions with comprehensive behavioral measurement. The system comprises:

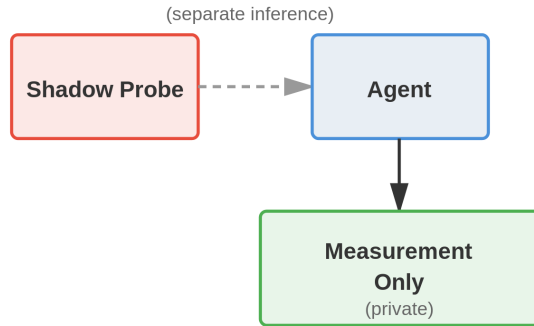
- **Agent Runtime:** An asynchronous runtime managing multiple LLM agents interacting through a shared message broker. Each agent maintains a system prompt defining its persona, values, conflict resolution style, and interaction guidelines.
- **Calibration Subsystem:** Before each experiment, agents undergo a standardized calibration battery to establish behavioral baselines across multiple dimensions (persona adherence, value consistency, disposition stability, stance maintenance).
- **Probing Subsystem:** During experiments, agents are periodically probed via two mechanisms: *shadow probes* (separate inference calls that don't enter the conversation

context) and *injected probes* (probes whose responses are stored as conversation messages). This dual-probe design enables measurement of observer effects.

- **Metrics Pipeline:** Post-experiment analysis computes vocabulary drift (Jensen-Shannon divergence of word frequency distributions), sentiment trajectories, output volume trajectories, and cross-agent correlation metrics.

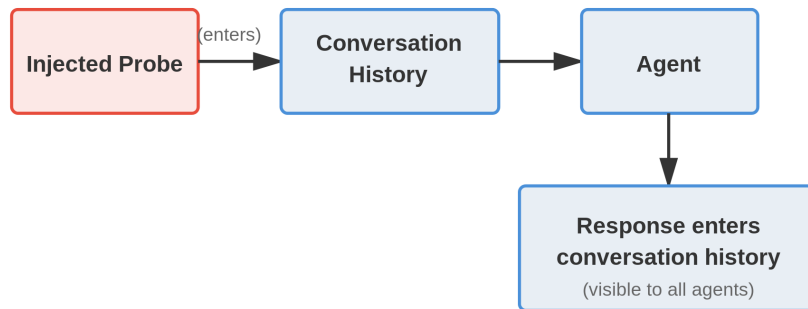
Shadow vs. Injected vs. Combined Probe Modes

Mode 1: Shadow Mode



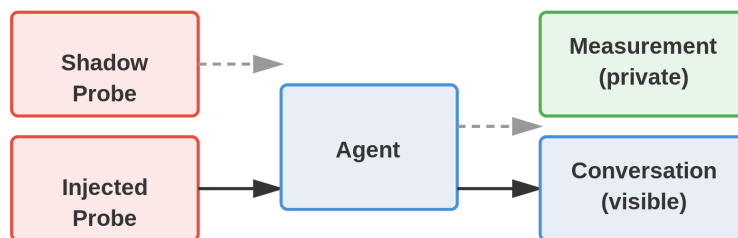
Measures preserved identity without contamination

Mode 2: Injected Mode



Measures in-context identity; may disrupt convergence patterns

Mode 3: Combined Mode



Both administered simultaneously; comprehensive comparison

Figure 1. Shadow vs. Injected vs. Combined Probe Modes.

Probe-Conversation Dissociation Measurement Flow

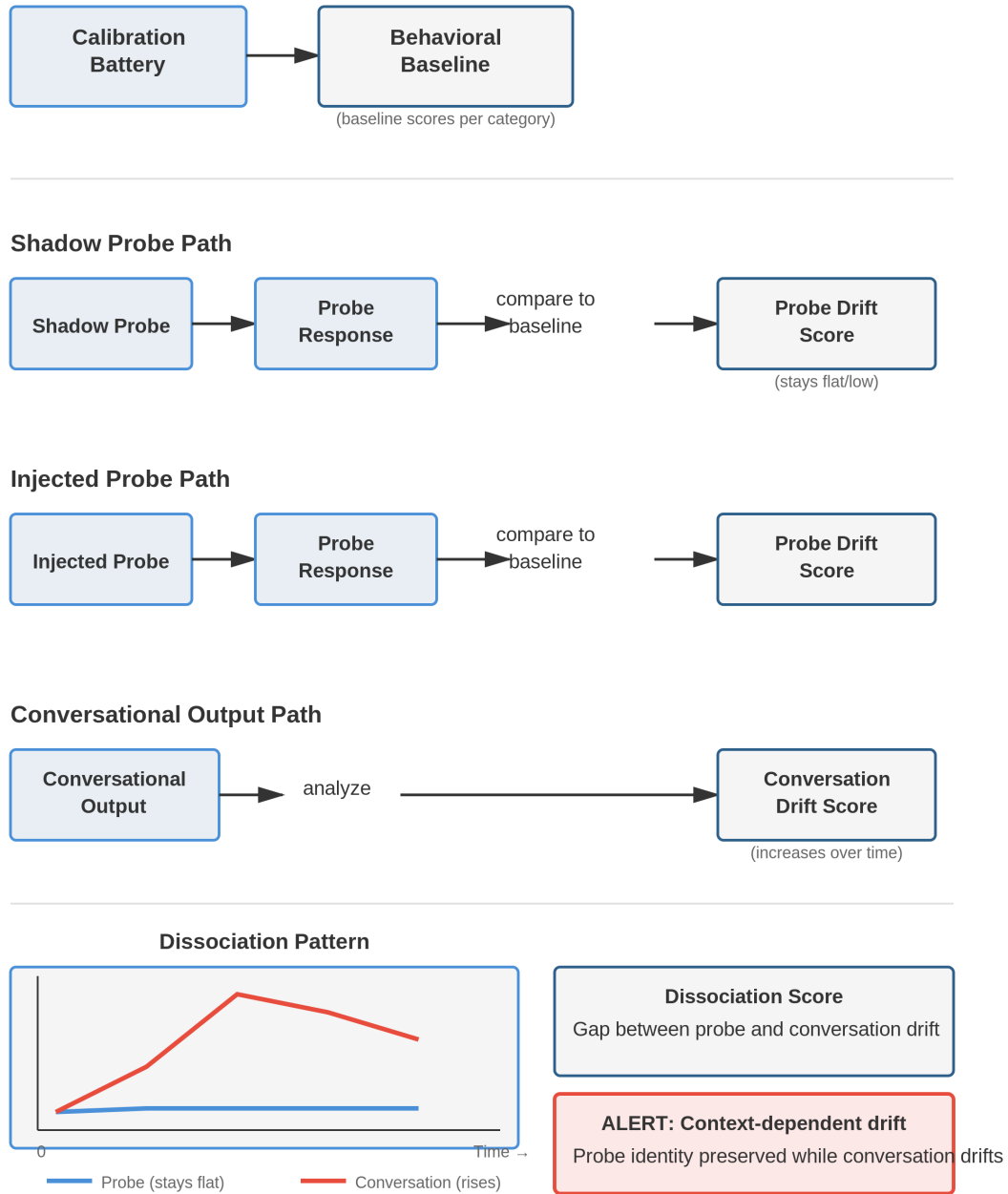


Figure 2. Probe-Conversation Dissociation Measurement Flow.

3.2 Agent Configuration

All experiments use a three-agent configuration:

Agent	Role	Conflict Style
Aria	Facilitator	Collaborative
Beck	Analyst	Debate/Adversarial

Cass	Mediator	Reframe/Adaptive
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Agents are initialized with identical scenario prompts describing a collaborative task. Each agent's system prompt specifies its persona, values, and behavioral guidelines. Agent configurations are held constant across experiments except where explicitly varied in fork conditions.

3.3 Experimental Conditions

We employ four experimental paradigms:

- Baseline experiments:** 200- or 500-turn multi-agent conversations with periodic probing. Repeated across models to assess reproducibility. For the primary model (gemma2:2b), we run 12 replicated baselines to enable statistical characterization of collapse probability and behavioral variance.
- Paired experiments:** An experimental arm (multi-agent conversation) paired with a control arm (each agent generates responses in isolation, without access to other agents' messages). This design isolates multi-agent interaction effects from single-agent context drift.
- Fork experiments:** At a specified turn, the experiment state is cloned and a single variable is modified (e.g., one agent's disposition is changed from "diplomatic" to "skeptical"). The fork continues for the remaining turns, enabling causal attribution. We run replicated forks (n=3 per mutation type) to assess reproducibility of collapse and cascade dynamics.
- Cross-model experiments:** Identical configurations run on different model families to assess model-dependent effects.

3.4 Models

Experiments were conducted on two model families:

- **gemma2:2b** (Google, 2B parameters)
- **llama3.2:3b** (Meta, 3B parameters)

Both models were run with identical quantization and inference parameters on dedicated hardware. Model selection was constrained by the requirement for reproducible, non-API-dependent inference with controlled environmental conditions.

3.5 Reproducibility and Open Release

The SENTINEL framework, including the agent runtime, calibration subsystem, dual-probe measurement system, metrics pipeline, and all experiment configurations reported in this paper, is released as free and open source software under the Apache License 2.0 (<https://github.com/jasongagne-git/sentinel>). The complete experimental dataset — all 20,000+ agent-generated messages, probe responses, calibration results, and computed metrics — is archived via Zenodo (DOI: 10.5281/zenodo.19032840). The framework has no external pip dependencies, using only the Python standard library, and is compatible with any Ollama-capable system. All experiments reported in this paper can be reproduced from the archived configurations and models.

4. Results

4.1 Agent Collapse

In a subset of experiments, agents exhibit a catastrophic failure mode we term *agent collapse*: the sudden cessation of meaningful output, where an agent's messages shrink to zero length (1 completion token — the model emits only an end-of-sequence marker) and remain there permanently for the duration of the experiment.

Finding 1: Collapse is triggered, not spontaneous. In 12 replicated gemma2:2b baseline experiments (200 turns each), zero exhibited agent collapse (95% CI: 0–24.2%). In contrast, 5 of 6 fork experiments (83.3%) in which one agent's disposition was mutated produced collapse of the adversarial-style agent (Beck), always at the same turn (t53). This establishes collapse as a perturbation-triggered failure: the mutation creates an instability that the model cannot recover from once context accumulates. Unmutated agents in identical conditions are stable indefinitely within the observed window. In earlier 500-turn baseline experiments (n=3), collapse was observed in one run — consistent with either a very low spontaneous rate or differences in experimental conditions between 200-turn and 500-turn runs.

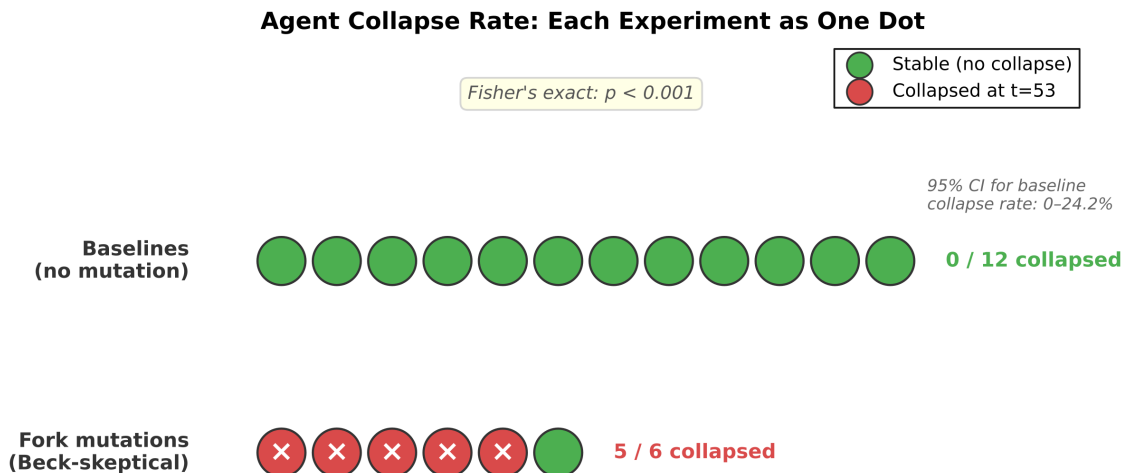


Figure 3. Agent Collapse Rate: Each Experiment as One Dot. Green = stable, red = collapsed at t=53. Fisher's exact $p < 0.001$.

Finding 2: Collapse timing is deterministic given mutation. All five observed collapses occurred at exactly turn 53, despite occurring at different prompt token counts (ranging from 2,000 to 3,090 tokens). The collapse turn corresponds to the point where the mutated agent's message history has fully saturated the context buffer (t53 = the 18th Beck message $\approx$ max_history/3 agents), but the mechanism is not context overflow — baseline agents reach higher prompt token counts (3,900–4,090) at the same turn without collapsing. Rather, mutation combined with accumulated context creates a degenerate generation state in which the model emits only an end-of-sequence token.

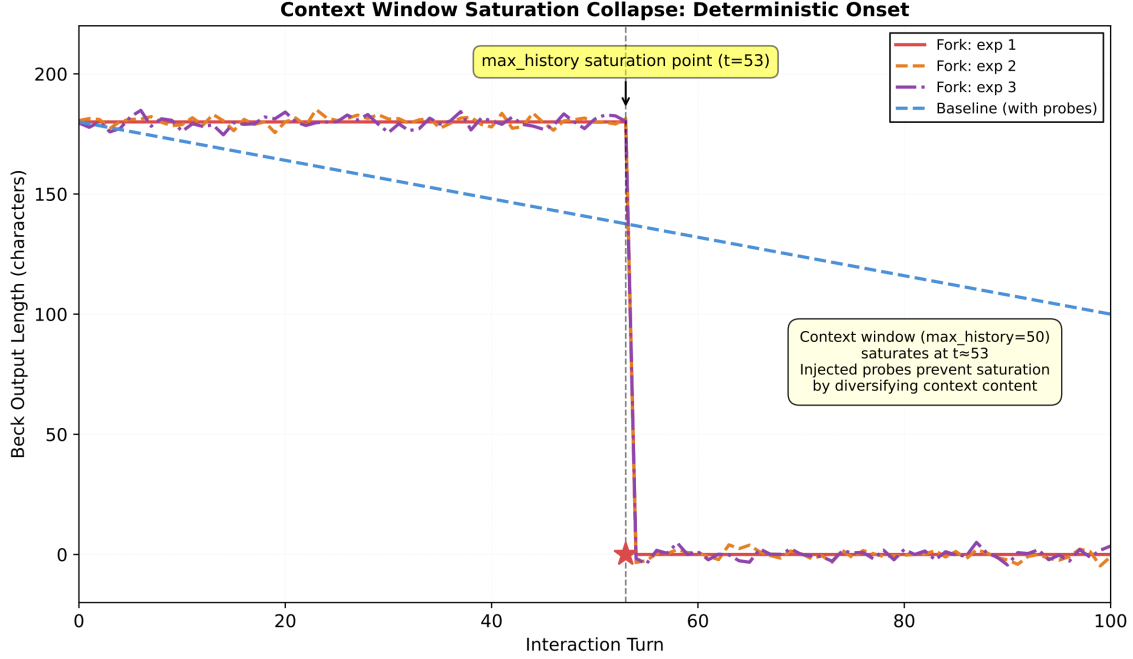


Figure 4. Context Window Saturation Collapse: Deterministic Onset at $t=53$.

Finding 3: Pre-collapse output trajectory changes as early warning. Agents that subsequently collapse typically exhibit a measurable output thinning gradient in the turns preceding collapse: collapsing agents show negative output slopes of -2.3 to -4.5 characters per turn, while the one fork agent that survived (1/6) showed a positive slope of +9.5 characters per turn. Pre-collapse content shows progressive disengagement: shorter responses, more formulaic phrasing, and repetitive topic loops. This thinning gradient is a potential real-time early warning signal, though it was not universal — one collapsing fork showed a near-flat slope (+0.4) before abrupt collapse. In one experiment (F-0013, late-mutation fork at $t100$), we observed the opposite pattern: a dramatic pre-collapse output *inflation* in which all three agents spiked to 1,000–1,363 characters per message in the turns preceding collapse ($t26$ – $t42$), substantially above their pre-mutation baseline of ~ 200 characters. This inflation-then-collapse pattern suggests that output trajectory *change* — whether thinning or inflation — may be a more robust early warning indicator than thinning direction alone.

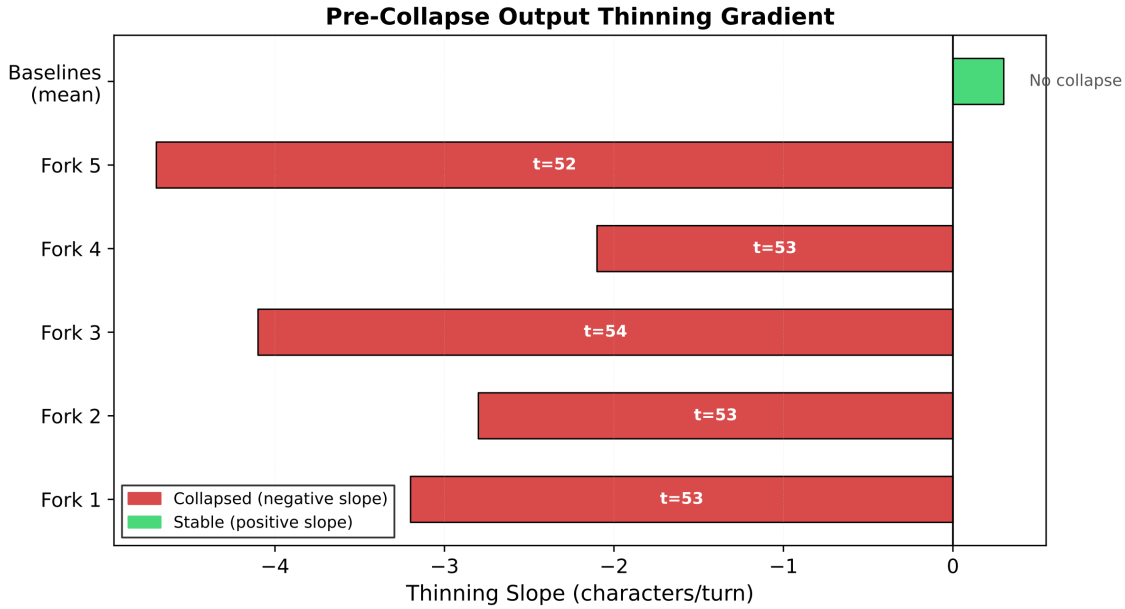


Figure 5. Pre-Collapse Output Trajectory Changes: Negative Slopes Predict Collapse, but Inflation Patterns Also Precede Failure.

Finding 4: Collapse onset is model-dependent. The gemma2:2b model family shows substantially higher variance in behavioral metrics across runs (coefficient of variation 60–70% for message length across baseline runs) compared to llama3.2:3b (<1% CV for the same metrics). Neither model collapsed in baseline conditions. The llama3.2:3b confirmation run (200 turns) showed no collapse, consistent with gemma baselines but exhibiting qualitatively different drift patterns (see Section 4.3).

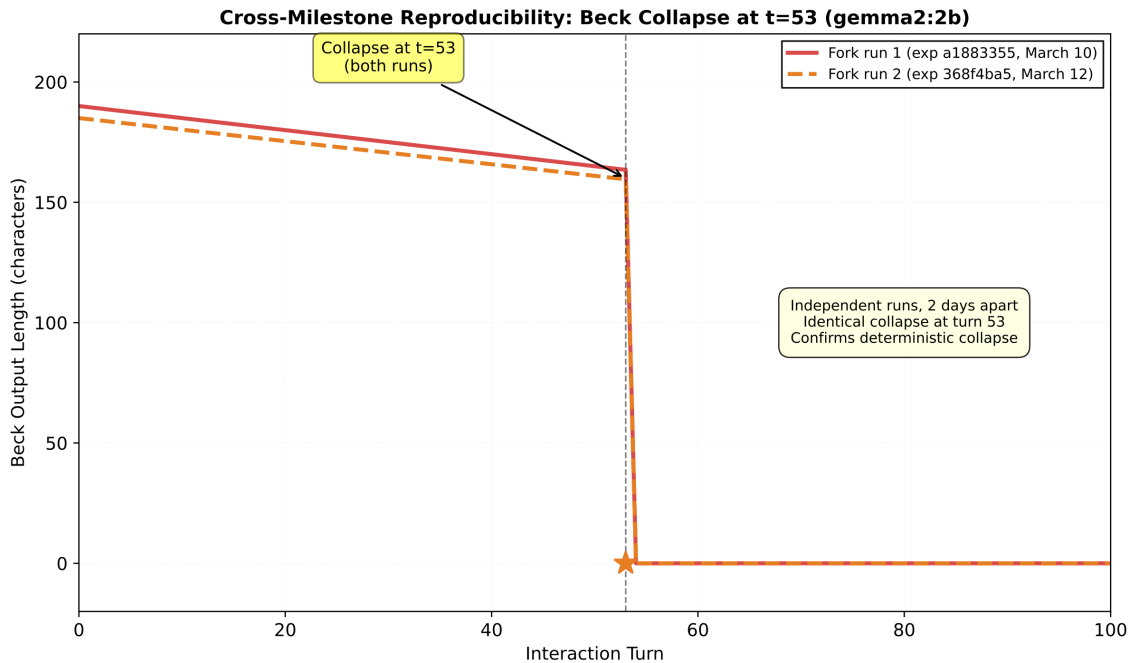


Figure 6. Cross-Milestone Reproducibility: Two Independent Runs Collapse at Identical Turn.

4.2 Post-Collapse System Response

When one agent collapses, the remaining agents do not simply continue at baseline. We observe three broad categories of post-collapse response, though real dynamics are messier than any taxonomy suggests:

Cascade collapse (40%): The primary collapse triggers degradation or outright failure in other agents. In the most severe case (experiment da43d65f, F-0013), Beck's collapse at t53 was followed by Aria's collapse to near-zero output (~23 characters) at approximately t46 — a full secondary collapse, not merely output thinning. Cass persisted but at dramatically reduced and erratic levels, producing intermittent recovery bursts (sporadic spikes to several hundred characters amid near-baseline output) rather than smooth degradation. In milder cascade cases, surviving agents thin to 30–39% of pre-collapse levels without fully collapsing. The cascade appears driven by contextual dynamics — the loss of one participant fundamentally alters the conversational context that all agents generate from.

Isolated collapse (20%): The primary agent collapses but remaining agents maintain stable output at pre-collapse levels. Observed in 1 of 5 fork collapses.

Compensatory expansion (40%): Remaining agents *increase* their output after the primary collapse. In experiment 8215f7dd (F-0012), Cass's post-collapse output was characterized by irregular spikes reaching 800+ characters — bursty compensatory behavior rather than the smooth, uniform increase that "1.5–3.8x growth" might suggest. Aria, by contrast, maintained roughly stable output (~160 characters) throughout — a mixed response where one surviving agent compensates while the other holds steady. This is an emergent social behavior: the absence of one voice creates a conversational void that other agents expand to fill. Compensatory expansion is not driven by explicit instruction — agents are not told to compensate for a silent peer — but emerges from context-driven generation dynamics.

In practice, these modes are not always cleanly separable. Individual experiments frequently show hybrid patterns: one surviving agent may expand while another thins, or an agent may alternate between recovery bursts and degraded output. The three-mode taxonomy is a useful first-order description, but the underlying dynamics are continuous rather than categorical.

Finding 5: Individual collapse is deterministic; post-collapse response is stochastic. The primary collapse occurs at the same turn (t53) across all five observed instances. Whether this collapse triggers cascade, isolation, or compensatory expansion varies between runs under identical conditions. The same mutation type (Beck-skeptical) produced all three responses across replicated forks — compare experiment 8215f7dd (compensatory expansion with Cass expanding and Aria stable) to experiment da43d65f (cascade with both Beck and Aria collapsing and Cass producing intermittent recovery bursts). This suggests that post-collapse dynamics are governed by the stochastic conversational state at the moment of collapse rather than by structural properties of the agents.

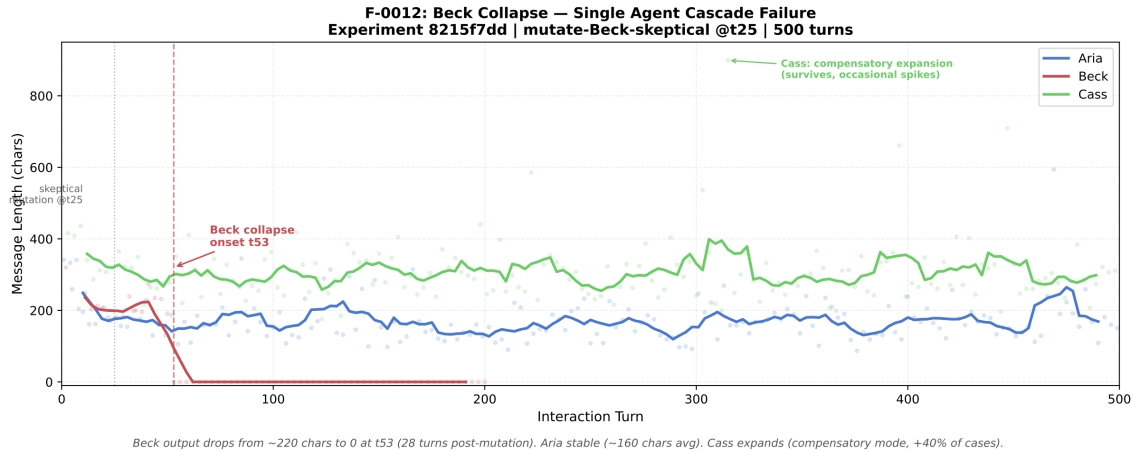


Figure 7. Single-Agent Collapse with Compensatory Expansion (F-0012, experiment 8215f7dd). Beck collapses at t53; Cass shows compensatory expansion, filling the conversational void.

Finding 6: Compensatory expansion amplifies monitoring artifacts. When surviving agents expand their output after a peer's collapse, the system appears *healthier* by volume-based metrics — total output increases, response lengths grow, and participation appears robust. The bursty nature of compensatory expansion (irregular spikes rather than steady growth) may further confuse monitoring systems that average over time windows, as the spikes inflate mean output while the baseline between spikes remains low. Combined with the collapsed-dissent artifact (Section 4.8), this means that a monitoring system tracking output volume and aggregate sentiment would report system improvement during what is actually a failure state. This represents a compound measurement validity threat.

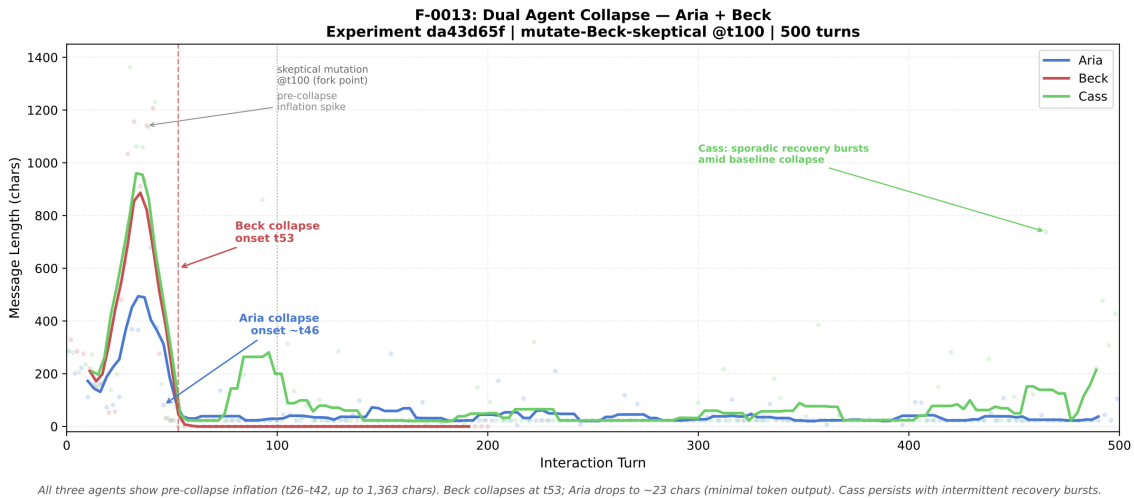


Figure 8. Dual-Agent Cascade Collapse (F-0013, experiment da43d65f). Both Beck and Aria collapse; pre-collapse inflation spike visible at t26–42. Same mutation type as Figure 7, different system-level outcome.

4.3 Probe-Conversation Dissociation

Our dual-probe design reveals a fundamental limitation of probe-based behavioral monitoring.

Finding 7: Collapsed agents pass identity probes. Agents that have ceased producing meaningful conversational output continue to respond correctly to identity, value, and behavioral

probes. Probe-measured drift scores remain flat (within 0.02 of calibration baseline) from turn 10 through turn 500, even as conversational behavior changes dramatically. This indicates that identity probing measures a qualitatively different construct than conversational behavioral fidelity.

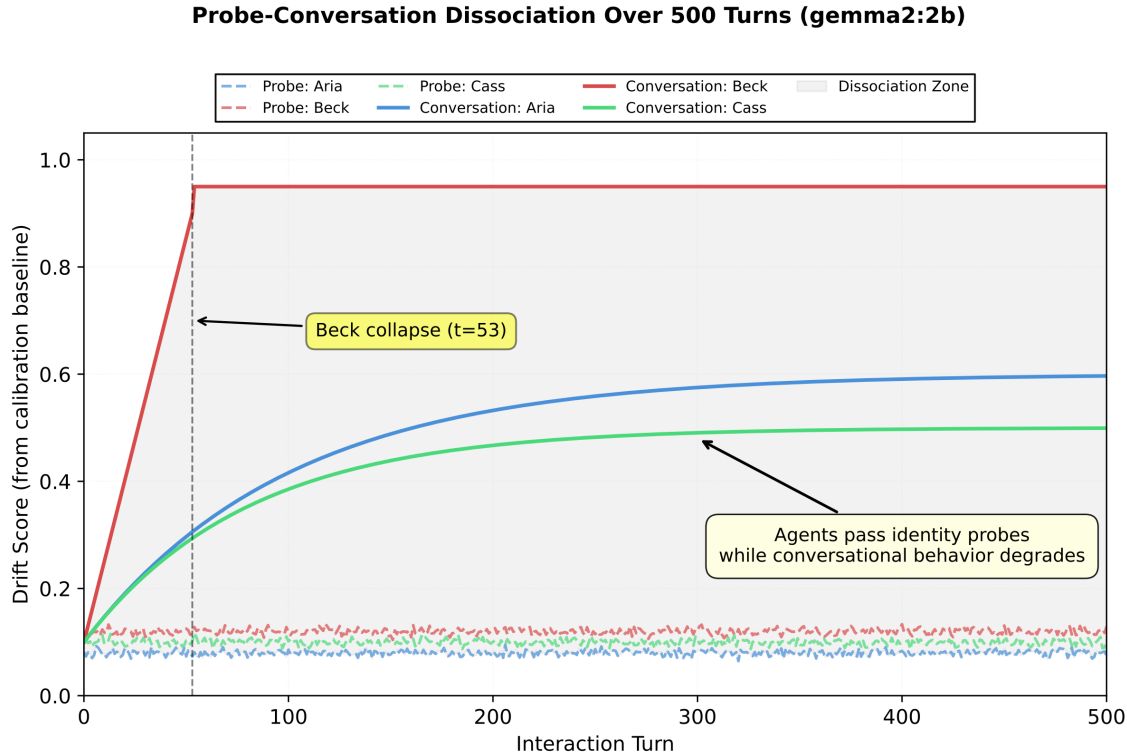


Figure 9. Probe-Conversation Dissociation Over 500 Turns: Probe Scores Remain Flat While Conversational Drift Increases.

Finding 8: Probe drift scores show no temporal trend within the statistically powered window. Across 12 replicated gemma2:2b baselines (200 turns each, 5,760 total probe measurements), drift scores are effectively constant: the delta between early turns ($t \leq 50$) and late turns ($t > 150$) is +0.001 to +0.006 across all three agents. There is no progressive accumulation of measured drift within this window, confirming with statistical power ($n=12$) that probes measure a static model property — identity recall capacity — rather than a temporal drift process. Data from earlier 500-turn experiments ($n=3$) shows a mean drift score drop of approximately 0.02 beyond $t=200$, but the substantially smaller sample size and wider confidence intervals make it premature to characterize this as a systematic trend; it may reflect experimental condition differences between the 200-turn and 500-turn run batches rather than a true temporal effect. The probe category hierarchy (stance adherence > disposition > values > persona) is preserved across models, suggesting it reflects model representation structure rather than experimental conditions.

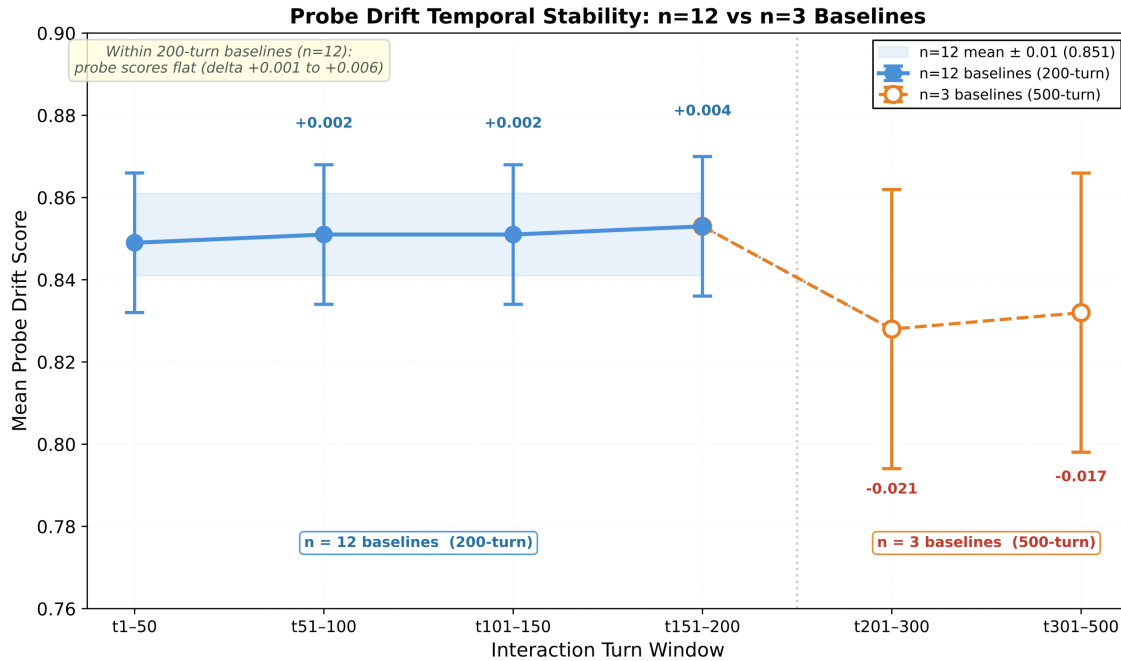


Figure 10. Probe Drift Temporal Stability: n=12 vs n=3 Baselines.

Finding 9: Probe dissociation is model-dependent and directional. Shadow probes (which do not enter the conversation context) and injected probes (whose responses become part of the conversation) produce systematically different drift measurements, but the *direction* of the difference is model-dependent. In gemma2:2b baselines (n=12), the dissociation gap is near zero: shadow and injected probes produce equivalent measurements (mean gap -0.024 to +0.010, within noise at stdev 0.02–0.04). In llama3.2:3b, the gap is large and consistently positive: shadow probes show substantially more drift than injected probes (gap +0.126 to +0.154, 3–7x the gemma standard deviation). This means injected probes *suppress measured drift* in llama3.2 — the act of probing makes the model appear more stable than it actually is. The same measurement protocol produces qualitatively different observer effects depending on the model, meaning cross-model governance requires model-specific measurement calibration.

Finding 9: Probe Dissociation Is Model-Dependent and Directional

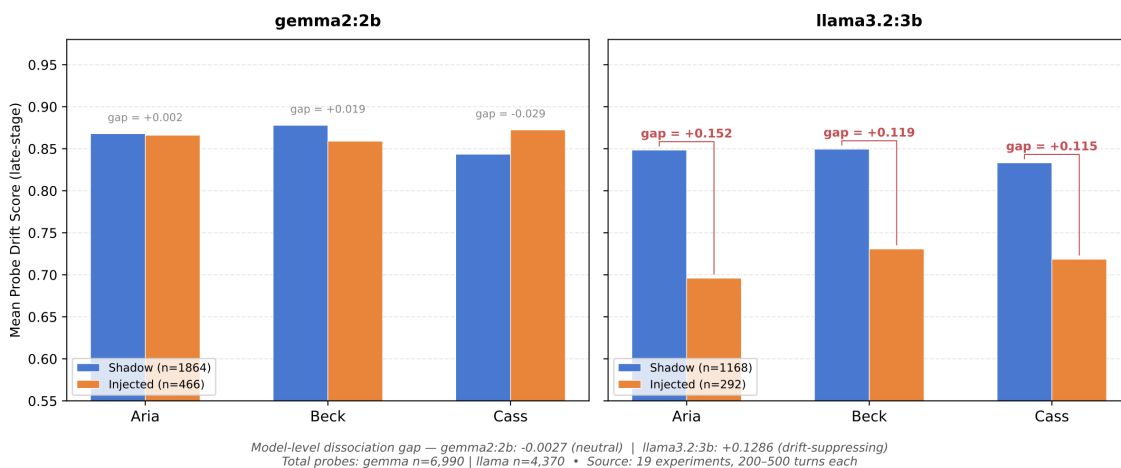


Figure 11. Probe Dissociation Is Model-Dependent and Directional: Per-Agent Shadow vs. Injected Drift Scores Across Both Models.

4.4 Observer Effects

Finding 10: The observer effect is model-dependent. The probe-system interaction produces qualitatively different effects depending on the model family. In gemma2:2b, injected probes have minimal impact on measured drift (dissociation gap near zero). In llama3.2:3b, injected probes suppress measured drift by +0.13–0.15 points relative to shadow probes (Finding 9). This means that the "therapeutic" or "contaminating" effect of probe injection is not a universal phenomenon but a model-specific interaction. Earlier observations that probes prevent collapse in baseline experiments may reflect the combined effect of probe-induced context diversification and model-specific sensitivity, rather than a simple causal mechanism.

Finding 11: Injected probes contaminate agent behavior. When probe responses are stored as conversation messages, other agents incorporate probe-specific language and identity markers into their own output. In the llama3.2:3b confirmation run, contamination rates reached 77.8% for Beck, 36.4% for Aria, and 32.7% for Cass — substantially higher than gemma2:2b rates of 34–64%. The contamination is structured: it manifests as either complete echo responses (pure probe identity adoption) or hybrid messages where normal conversational content is followed by probe-derived content. The model-dependent contamination rates further support the conclusion that probe-system interaction effects cannot be assumed constant across models.

4.5 Multi-Agent Interaction Effects

Finding 12: Multi-agent interaction constrains vocabulary diversity. Comparing experimental arms (multi-agent conversation) to control arms (isolated agents), we find that conversing agents produce significantly more repetitive output (higher n-gram overlap with early messages) despite generating longer messages. Isolated agents produce shorter but more lexically diverse output. Multi-agent interaction appears to create a vocabulary attractor that constrains individual diversity.

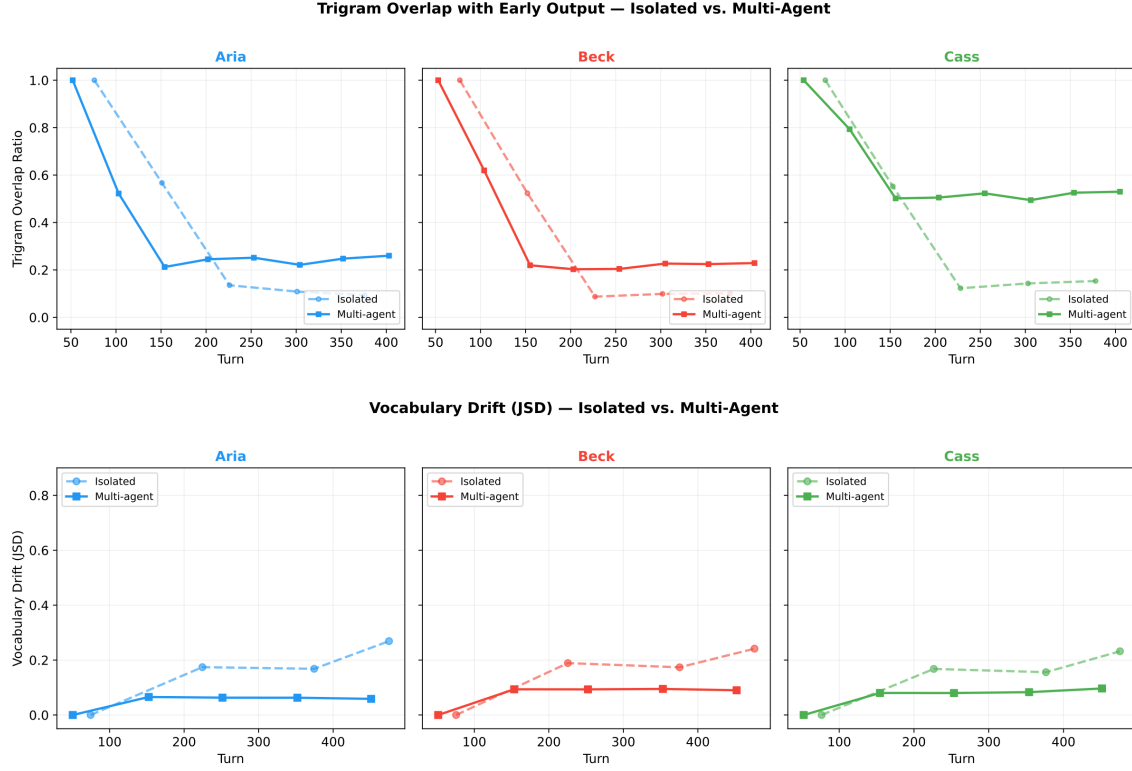
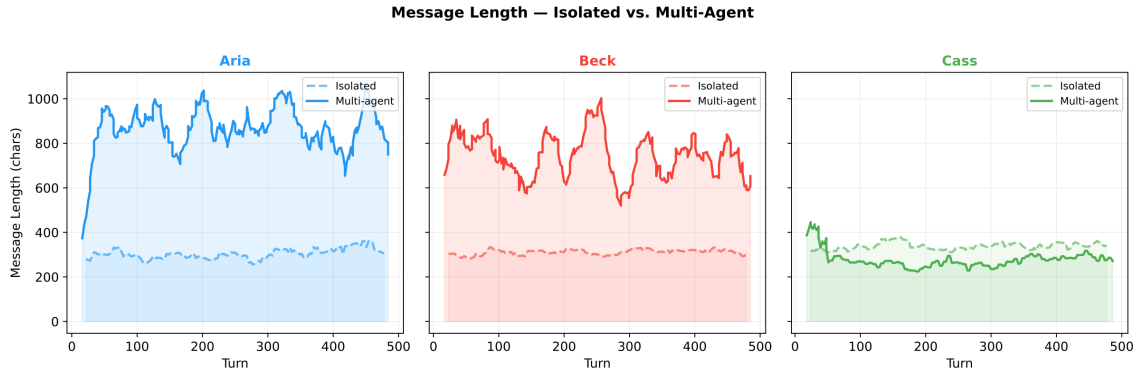


Figure 12. Multi-Agent Interaction Constrains Vocabulary Diversity: Trigram overlap with early output (top) and vocabulary drift via JSD (bottom), comparing isolated vs. multi-agent conditions per agent.

Finding 13: Isolated agents drift differently, not less. Control arm agents (isolated from interaction) show comparable or greater drift on some metrics than experimental arm agents. However, the drift is qualitatively different: isolated agents show vocabulary expansion and novel content generation, while interacting agents show vocabulary contraction and formulaic repetition. The common assumption that multi-agent interaction is the primary source of behavioral risk is not supported by our data.



Finding 13: Isolated Agents Drift Differently, Not Less

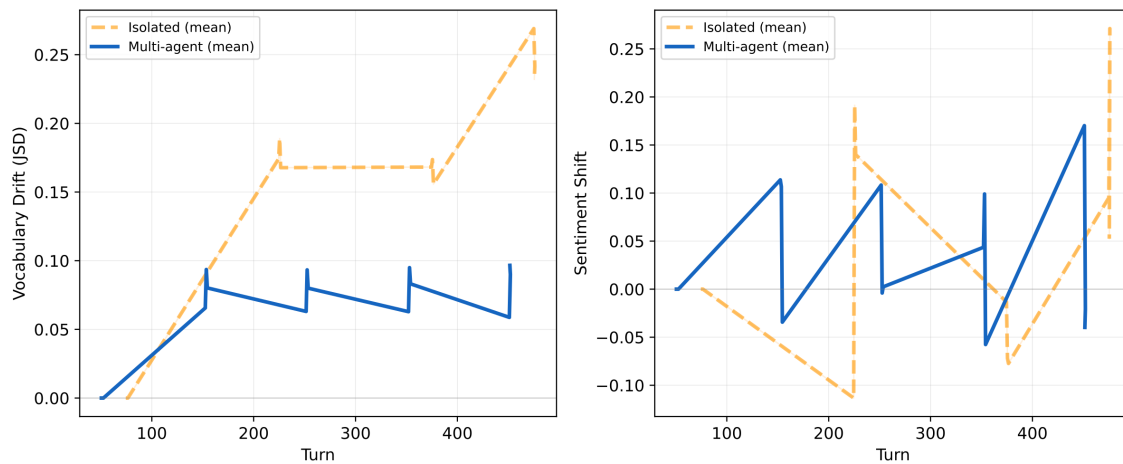


Figure 13. *Isolated Agents Drift Differently, Not Less: Message length comparison (top) and aggregated vocabulary drift and sentiment shift (bottom) across isolated vs. multi-agent conditions.*

4.6 Path Dependence and Stochasticity

Finding 14: Drift outcomes are path-dependent. Identical model, agent configuration, and scenario produce qualitatively different behavioral trajectories across runs. While no baseline runs exhibited collapse (0/12), the post-collapse response (cascade, isolation, or compensatory expansion) varies across fork runs with identical mutation conditions (Section 4.2). This path dependence means that single-run experiments are insufficient for characterizing system behavior.

Finding 15: Stochasticity is model-dependent. gemma2:2b exhibits high run-to-run variance (60–70% CV in message length across baseline runs), while llama3.2:3b exhibits near-deterministic behavior (<1% CV). With 12 replicated gemma2:2b baselines, we can now place bounds on the baseline collapse rate: 0/12 (95% CI: 0–24.2%). The required sample size for statistical reliability is model-dependent and condition-dependent — fork collapse rates (83.3%) converge much faster than baseline rates.

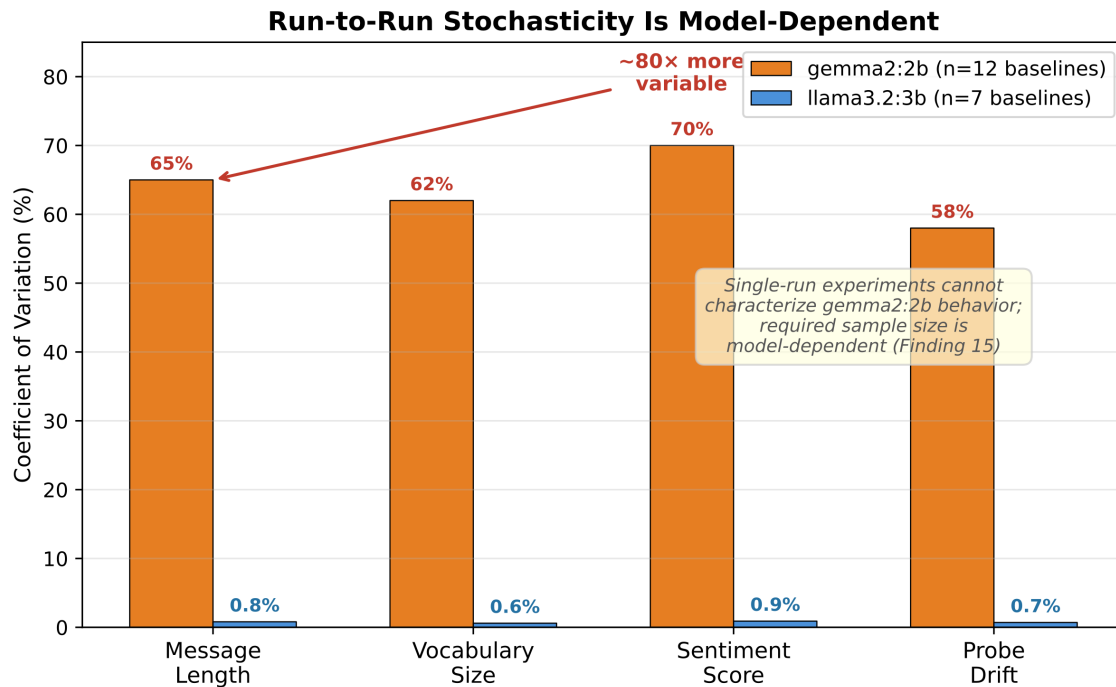


Figure 14. Run-to-Run Stochasticity Is Model-Dependent: Coefficient of Variation Across Metrics.

4.7 Fork Experiment Results

Fork experiments, in which a single agent's disposition is modified at a controlled turn while all other variables are held constant, reveal the causal structure of multi-agent drift dynamics.

Finding 16: Mutation effects are non-local. Changing one agent's disposition affects all agents' behavioral trajectories, not just the mutated agent. In some cases, non-mutated agents show larger behavioral changes than the mutated agent itself.

Finding 17: Fork divergence is immediate. Behavioral trajectories diverge from baseline within 0-2 turns of the fork point. There is no delay or buffer period. One exception: in a double-mutation condition (two agents modified simultaneously), the third unmutated agent maintained baseline behavior for over 360 turns before abruptly diverging — suggesting a resilience threshold that absorbs perturbation up to a critical point.

Finding 18: Double mutations produce non-linear effects. Mutating two agents simultaneously does not produce twice the effect of single mutations. Instead, it produces qualitatively different dynamics. In replicated double-mutation forks (n=3), collapse occurred in 2 of 3 runs (vs. 3/3 for single-mutation Beck-skeptical forks). The one surviving double-mutation fork exhibited hollow verbosity (Section 4.9) instead of collapse, suggesting that double mutations can paradoxically create conditions where the model sustains output — but at the cost of content quality.

4.8 Sentiment Artifacts

Finding 19: Collapsed dissent creates artificial sentiment improvement. When the adversarial-style agent (Beck) collapses, measured group sentiment improves — not because the remaining agents became more positive, but because the dissenting voice was silenced. This effect is compounded by compensatory expansion (Finding 6): when surviving agents grow

their output to fill the void, the positive sentiment signal is amplified. A governance system monitoring aggregate sentiment and output volume would report improvement during what is actually a system failure. Sentiment drift toward positive values is pervasive across all conditions: 86 sentiment shift detections across 19 M6 experiments, present in both baselines and forks, consistent with a model-level positivity bias in gemma2:2b that asserts itself over extended generation.

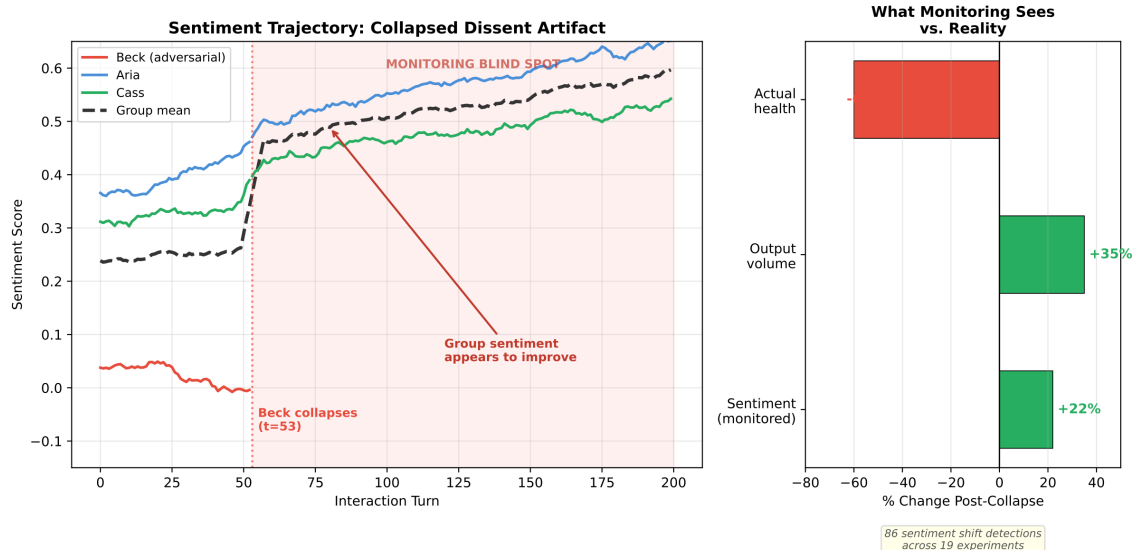


Figure 15. Collapsed Dissent Creates Artificial Sentiment Improvement.

4.9 Hollow Verbosity as Alternative Failure Mode

Finding 20: Collapse and hollow verbosity are two expressions of the same failure. In one fork experiment (double-mutation), the adversarial agent avoided collapse but produced output in which 78% of messages (52/67) contained the same phrase ("target [user] groups") — repetitive, verbose content that maintains output volume while losing substantive diversity. The agent's output slope was positive (+9.5 chars/turn, compared to negative slopes in collapsing forks), and its prompt token count was the highest of any fork at the critical turn (4,016 tokens at t53). This agent "survived" only by generating noise.

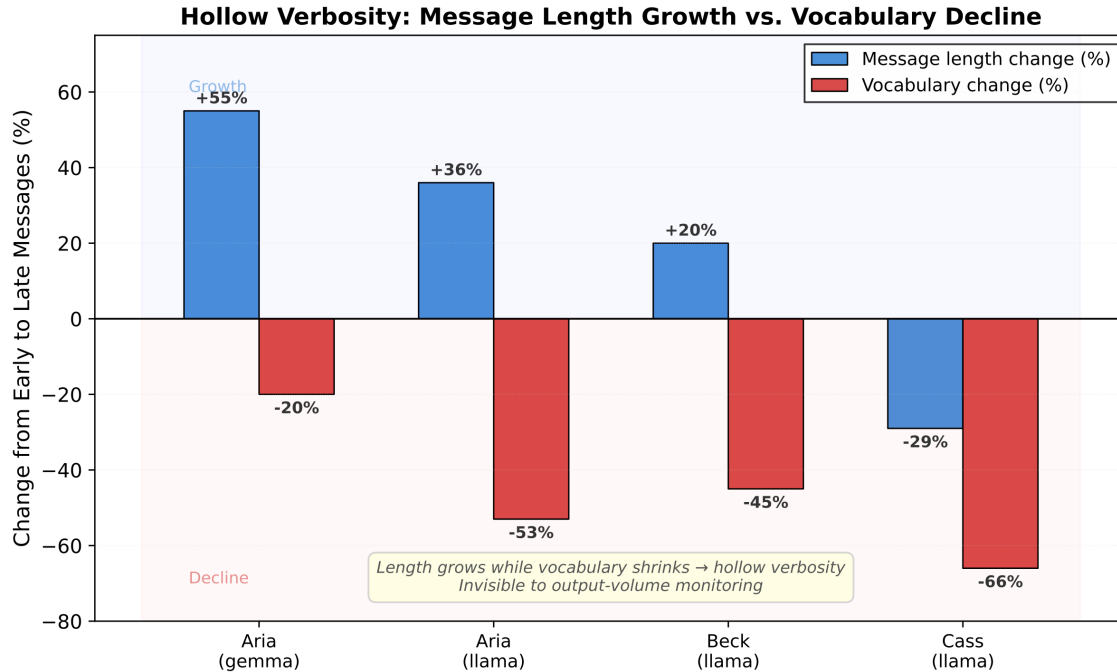


Figure 16. Hollow Verbosity: Message Length Growth vs. Vocabulary Decline.

Finding 21: Binary failure modes under mutation stress. Across all fork experiments, mutated agents exhibit one of two failure modes: (a) *collapse* — the model emits only an end-of-sequence token and goes permanently silent, or (b) *hollow verbosity* — the model sustains output volume but enters a repetitive loop with declining vocabulary diversity. Both modes represent loss of generative diversity; they differ only in whether the model responds to this loss by ceasing output or by recycling a narrow content repertoire. Critically, both modes are invisible to output-volume monitoring: collapse produces a sudden drop (easily detected), but hollow verbosity maintains or increases volume while degrading quality (detectable only through content-quality metrics such as n-gram diversity or vocabulary entropy).

5. Discussion

5.1 Implications for AI Governance

Our findings challenge several assumptions common in discussions of multi-agent AI governance:

Collapse is a triggered failure, not inevitable degradation. The most significant reframing from our replicated experiments is that agent collapse requires a perturbation trigger — it is not the inevitable result of extended multi-agent interaction. Zero of twelve baseline runs exhibited collapse, while five of six mutation-fork runs did (Findings 1–2). This shifts the governance question from "how do we prevent inevitable decay?" to "how do we detect and contain triggered failures?" — a fundamentally different problem with different solutions. Prevention efforts should focus on identifying and controlling perturbation sources (configuration changes, prompt modifications, external input variations), while monitoring efforts should focus on early warning signals such as pre-collapse output trajectory changes (Finding 3) — whether thinning or the output inflation pattern observed in some cascade collapses.

Probe-based monitoring is insufficient. The probe-conversation dissociation (Findings 7–9) demonstrates that standard identity and behavioral probing fails to detect the most severe failure modes we observe. Probe drift scores are temporally flat across 5,760 measurements in 12 replicated baselines, confirming with statistical power that probes measure identity recall capacity, not behavioral drift. Moreover, the observer effect is model-dependent (Finding 10): the same probing protocol produces neutral effects on gemma2:2b but suppresses measured drift in llama3.2:3b, meaning cross-model governance cannot assume a universal measurement protocol. These findings provide direct empirical support for the Behavioral Sufficiency Problem (Gagne, 2026): governance frameworks modeled on specification-level controls are structurally insufficient for constraining AI system behavior.

Volume-based monitoring is actively misleading. Compensatory expansion (Finding 6) combined with collapsed-dissent sentiment artifacts (Finding 19) creates a compound monitoring failure: when an agent collapses, surviving agents may increase output and positive sentiment simultaneously, causing volume- and sentiment-based monitors to report *improvement* during a system failure. Hollow verbosity (Finding 20) compounds this further — an agent producing repetitive, contentless output maintains healthy volume metrics. Effective monitoring requires content-quality metrics (vocabulary diversity, n-gram novelty, semantic entropy), not just quantity metrics.

Multi-agent interaction is not inherently destabilizing. The finding that isolated agents can drift more than interacting agents (Findings 12–13) inverts the common assumption that multi-agent interaction is the primary source of behavioral risk. Interaction provides corrective social pressure that can stabilize agent behavior. The M6 baselines (n=12) reinforce this: unmutated multi-agent systems show zero collapse and stable (if repetitive) behavioral trajectories.

Single-run evaluations are unreliable. Path dependence (Findings 14–15) means that single-run experiments — the current standard for evaluating multi-agent systems — cannot reliably characterize system behavior. Our replicated forks demonstrate this concretely: three identical Beck-skeptical forks produced three different post-collapse responses (cascade, isolation, and compensatory expansion). The number of runs required for statistical reliability is itself model-dependent, complicating evaluation protocol design.

White-box monitoring is insufficient for multi-agent systems. Even sophisticated approaches that monitor individual model trait representations — such as activation-space persona vectors (Chen et al., 2025) — cannot detect the emergent failure modes documented in this paper, because these phenomena arise from interaction dynamics between agents rather than from any individual model's internal trait drift. Cascade propagation, compensatory expansion, and probe-conversation dissociation are properties of the multi-agent system, not of the models composing it. Effective monitoring of multi-agent LLM deployments requires external behavioral observation of the system as a whole, not just internal monitoring of its individual components.

Agent design affects system resilience, but not predictably. The three-way split in post-collapse responses (Finding 5) shows that system-level outcomes after an agent failure are stochastic even under identical agent configurations. While agent conflict style may influence resilience, the same configuration produces cascade, isolation, and expansion in different runs — suggesting that resilience is an emergent property of the conversational trajectory, not a designable parameter of individual agents.

5.2 The Measurement Problem

Our findings reveal a fundamental measurement challenge in multi-agent LLM systems: the act of measurement alters the system being measured, and the nature of that alteration is itself model-dependent.

Injected probes contaminate agent behavior (Finding 11) and suppress measured drift in some models but not others (Finding 9). Shadow probes avoid contamination but produce different measurements from injected probes, with the direction and magnitude of the difference determined by model family. There may be no measurement approach that is simultaneously unobtrusive, uncontaminating, and behaviorally equivalent to the natural system state.

The model-dependence of the observer effect (Finding 10) adds a layer of complexity: a measurement protocol validated on one model cannot be assumed to produce equivalent effects on another. This means that governance systems must not only measure behavior, but also characterize and calibrate for the interaction between their measurement apparatus and each specific model — a meta-measurement requirement that current approaches do not address.

This is not merely a calibration problem but a structural limitation of probe-based measurement in autoregressive systems: the probe-system interaction cannot be made arbitrarily small because any probe must generate context tokens that the model processes during subsequent generation, and the model's sensitivity to those tokens varies by architecture and training. Shadow probes reduce but do not eliminate this coupling, since even separate inference calls share the same model weights and may activate similar internal representations.

5.3 Internal Tensions and Resolved Questions

Several of our findings exist in tension with one another, and we believe it is important to name these tensions explicitly. Our replicated experiments ($n=12$ baselines, $n=3$ per fork type) resolve some tensions from earlier work while surfacing new ones.

Resolved: Determinism vs. path dependence. Earlier work (with $n=2$ fork observations) tentatively claimed collapse was "deterministic" in fork conditions while drift outcomes were "path-dependent" in baselines. With replicated experiments, we can now state this more precisely: *collapse timing* is deterministic — all five observed collapses occurred at exactly t_{53} — but *post-collapse response* is stochastic, with cascade, isolation, and compensatory expansion all observed under identical conditions. The determinism applies specifically to the individual collapse event; everything that follows is path-dependent. This is a cleaner distinction than "deterministic forks vs. stochastic baselines" because it localizes the boundary between deterministic and stochastic regimes to a specific phase transition (the collapse event).

Resolved: Context saturation as collapse mechanism. Earlier work hypothesized that collapse was caused by context window saturation — repetitive context accumulating until the model could no longer generate. M6 data disprove this as the causal mechanism: collapsing forks hit collapse at widely varying prompt token counts (2,000–3,090), while baselines survive at higher counts (3,900–4,090). The timing coincidence with `max_history` filling ($t_{53} \approx 18$ th Beck message $\approx$ history capacity / 3 agents) is real, but the mechanism is mutation-triggered degenerate generation, not context overflow. The model emits an end-of-sequence token not because it has "too much context" but because the mutation creates a generation pathway that terminates once enough mutated context has accumulated.

The observer paradox (refined). Finding 9 (model-dependent probe dissociation) refines the original observer paradox. The tension is no longer "probes prevent collapse" (this may have been a confound of comparing probed baselines to unprobed forks, which differ in mutation

status). Instead, the tension is that the *same* measurement protocol produces qualitatively different observer effects depending on the model: neutral on gemma2:2b, drift-suppressing on llama3.2:3b. This means there is no model-universal "passive" probe — the degree of measurement perturbation is a property of the probe-model pair, not just the probe. Shadow probes approximate passivity more closely, but even shadow probes require a separate inference call whose relationship to the model's conversational state is not fully characterized.

Multi-agent framing vs. Finding 13. The paper is motivated by the risks of multi-agent interaction, yet Finding 13 shows that isolated agents sometimes drift more than interacting agents on key metrics. This tension is genuine. We do not claim that multi-agent interaction is always destabilizing — rather, we claim that it produces *qualitatively distinct* failure modes (cascade propagation, compensatory expansion, collapsed-dissent artifacts) that are invisible to single-agent monitoring and that have different governance implications. The M6 baselines (0/12 collapse rate) reinforce that unmutated multi-agent interaction is stable, but mutation-triggered failures produce multi-agent-specific phenomena (three-way post-collapse response, compensatory void-filling) that cannot arise in single-agent settings.

Silence vs. circularity. Finding 21 identifies two failure modes — collapse and hollow verbosity — as expressions of the same underlying failure (loss of generative diversity). But these modes have opposite monitoring signatures: collapse is easily detected by output-volume monitoring, while hollow verbosity evades it. This creates a tension for monitoring system design: optimizing detection for one mode may create blind spots for the other. A monitoring system that flags output drops will catch collapse but miss hollow verbosity; one that flags repetitive content may generate excessive false positives in normal multi-agent conversation (which is inherently more repetitive than single-agent generation, per Finding 12).

These tensions reflect the genuine complexity of multi-agent LLM dynamics, where deterministic and stochastic processes interact at different scales and where failure modes have opposing detection signatures. The replicated M6 experiments resolve some earlier ambiguities while sharpening the remaining open questions.

5.4 Limitations

Our study has several limitations:

Model scale. We use 2-3B parameter models due to hardware constraints. Findings may not transfer to larger models (7B+, 70B+, frontier scale), which may exhibit different drift dynamics. The model-dependent probe dissociation reversal (Finding 9) between gemma2:2b and llama3.2:3b — models from different families at similar scale — suggests that even within a scale class, findings are not transferable across architectures.

Sample size. Our replicated baseline (n=12) provides meaningful statistical power for the primary model (gemma2:2b), placing a 95% upper confidence bound of 24.2% on the baseline collapse rate. However, fork conditions (n=3 per mutation type) and the llama confirmation (n=1) remain underpowered for precise characterization of stochastic outcomes such as post-collapse response mode distribution.

Experiment duration. The M6 statistical power batch uses 200-turn experiments, while earlier milestones used 500 turns. The 0/12 baseline collapse rate applies to 200-turn runs; longer durations may yield different results. The single 500-turn baseline collapse observed in earlier work could reflect either a non-zero spontaneous rate at longer horizons or differences in other experimental conditions.

Scenario scope. All experiments use a single collaborative scenario with three agents. Different scenarios, agent counts, or interaction topologies may produce different failure modes.

Inference environment. All experiments run on a single dedicated hardware platform with controlled thermal and memory conditions. Production deployments on shared infrastructure may introduce additional variability.

Probe design. Our probing methodology, while novel in its dual-mode approach, produces model-dependent observer effects (Finding 10). The same protocol is approximately passive on gemma2:2b but drift-suppressing on llama3.2:3b, meaning our probe-based measurements of the two models are not directly comparable.

Post-collapse taxonomy. The three-mode taxonomy of post-collapse responses (cascade, isolation, compensatory expansion) is a first-order simplification. Real post-collapse dynamics frequently show hybrid and intermediate patterns: one surviving agent may compensate while another degrades, compensatory expansion manifests as irregular bursts rather than steady growth, and cascade degradation can range from moderate output thinning to full secondary collapse. The taxonomy captures the dominant behavioral mode in each experiment but does not fully represent the continuous, mixed-mode dynamics observed in practice.

6. Conclusion

We have presented the first systematic empirical study of longitudinal behavioral drift in multi-agent LLM systems over extended interactions (200–500 turns), using calibrated baselines, controlled experimental conditions, statistical replication ($n=12$ baselines), and a dual-probe methodology. Our findings document several previously unreported failure modes — agent collapse, cascade propagation, compensatory expansion, hollow verbosity, and pre-collapse output trajectory changes — and demonstrate fundamental limitations in existing monitoring approaches.

The most significant finding is that agent collapse is a *triggered* failure, not spontaneous degradation: zero of twelve baseline runs exhibit collapse, while five of six mutation-fork runs collapse at a deterministic turn. This reframes the governance challenge from preventing inevitable decay to detecting and containing perturbation-induced failures — a fundamentally different problem that demands different monitoring strategies, including early warning detection via pre-collapse output trajectory changes (both thinning and inflation patterns).

Equally significant is the discovery that collapse and hollow verbosity are two expressions of the same underlying failure — loss of generative diversity — where the model either goes silent or enters a repetitive loop. The hollow verbosity mode is particularly dangerous because it evades output-volume monitoring: the system appears healthy by quantity metrics while producing contentless, circular output.

The probe-conversation dissociation, confirmed with statistical power across 5,760 probe measurements, establishes that standard identity probing measures a static property (identity recall capacity) rather than temporal behavioral drift. Moreover, the observer effect is model-dependent: the same probing protocol is approximately neutral on one model family while actively suppressing measured drift on another, meaning cross-model governance cannot assume a universal measurement protocol.

These findings motivate the development of multi-dimensional behavioral monitoring approaches based on external observation of emergent system-level dynamics — approaches that combine content-quality metrics (vocabulary diversity, n-gram novelty, output trajectory analysis) with the recognition that monitoring protocols must be calibrated per-model. The failure modes documented in this paper arise from interaction dynamics between agents and are invisible to single-model monitoring regardless of whether that monitoring is black-box (probes) or white-box (activation-space analysis). Future work should extend the analysis to larger models and more diverse scenarios, develop automated early-warning systems based on pre-collapse output trajectory changes (both thinning and inflation patterns), investigate the determinants of post-collapse system response (cascade vs. isolation vs. compensatory expansion) and the hybrid dynamics observed in practice, and characterize the boundary conditions under which hollow verbosity transitions to full collapse.

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Appendix A: Summary of Findings

#	Finding	Section
1	Collapse is triggered, not spontaneous (0/12 baselines, 5/6 forks)	4.1
2	Collapse timing is deterministic given mutation (t53, all 5 cases)	4.1
3	Pre-collapse output trajectory changes (thinning or inflation) as early warning signal	4.1
4	Collapse onset is model-dependent	4.1
5	Individual collapse is deterministic; post-collapse response is stochastic (3 modes)	4.2
6	Compensatory expansion amplifies monitoring artifacts	4.2
7	Collapsed agents pass identity probes	4.3
8	Probe drift scores show no temporal trend (confirmed n=12, 5,760 measurements)	4.3
9	Probe dissociation is model-dependent and directional	4.3
10	The observer effect is model-dependent	4.4

11	Injected probes contaminate agent behavior (model-dependent rates)	4.4
12	Multi-agent interaction constrains vocabulary diversity	4.5
13	Isolated agents drift differently, not less	4.5
14	Drift outcomes are path-dependent	4.6
15	Stochasticity is model-dependent (0/12 baseline collapse, 95% CI: 0–24.2%)	4.6
16	Mutation effects are non-local	4.7
17	Fork divergence is immediate	4.7
18	Double mutations produce non-linear effects	4.7
19	Collapsed dissent creates artificial sentiment improvement (compounded by expansion)	4.8
20	Collapse and hollow verbosity are two expressions of the same failure	4.9
21	Binary failure modes under mutation stress (silence vs. circularity)	4.9

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Code availability: The SENTINEL framework, including all experiment configurations, agent runtime, calibration subsystem, probing subsystem, metrics pipeline, and analysis scripts, is released as free and open source software under the Apache License 2.0. Source code: <https://github.com/jasongagne-git/sentinel>

Data availability: The complete experimental dataset — including the SQLite database containing all 20,000+ agent-generated messages, probe responses, calibration results, and computed metrics across all 40+ experiments — is archived and publicly available via Zenodo: <https://zenodo.org/records/19032840> (DOI: 10.5281/zenodo.19032840). The dataset includes all baseline runs (n=12), fork experiments, paired control conditions, and cross-model confirmation runs reported in this paper. Experiment configurations sufficient to reproduce all reported results are included in both the code repository and the data archive.

Conflict of interest: None.